

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2-3 Marks)	Level 1 – Beginning (0-1 Mark)
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Q. No	Understanding of Problem Context (20)	Application of Concepts (30)	Problem-Solving Approach (20)	Accuracy of Solution (20)	Clarity (10)	Total (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 20	/ 30	/ 20	/ 20	/ 10	/ 100

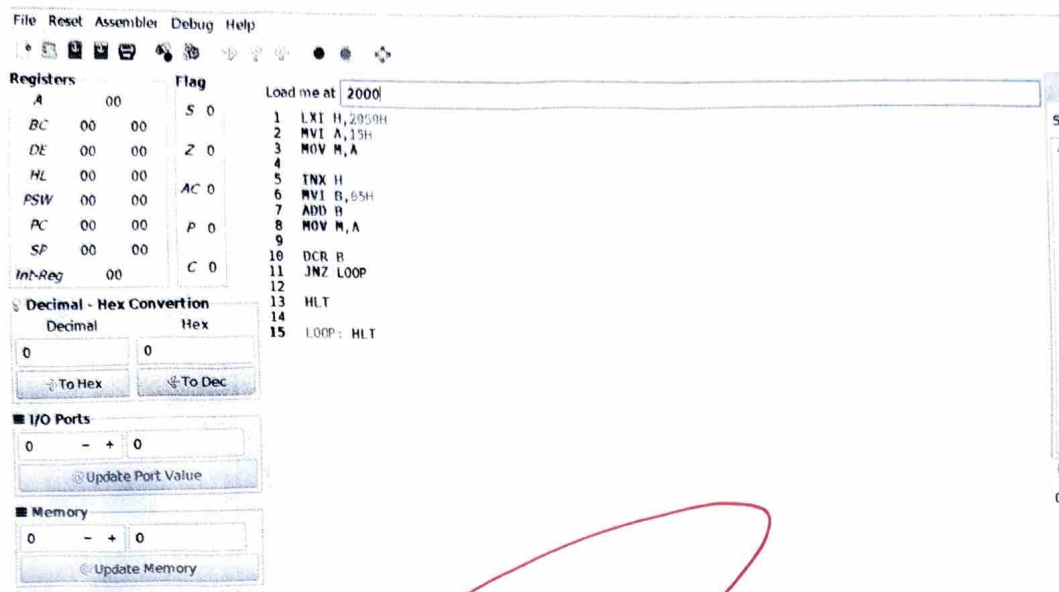
Course Coordinator

Faculty Incharge

Q1. 8085 Microprocessor Program Execution

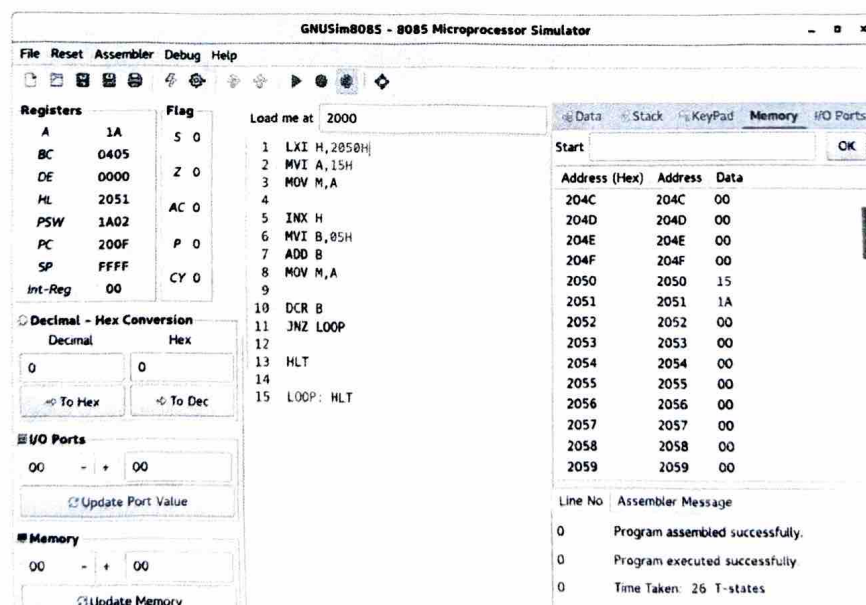
Objective

To study the execution of data transfer, arithmetic, and branch instructions in the 8085 microprocessor.



Explanation:

- The program stores 15H at memory location 2050H.
- It adds 05H to the accumulator and stores the result at 2051H.
- JNZ demonstrates branch instruction execution.



Observation:

- Accumulator (A) = 1AH
- Register B = 04H
- Memory 2050H = 15H
- Memory 2051H = 1AH
- Program execution completed successfully.

Q2. Bitwise Operations Using Windows Calculator

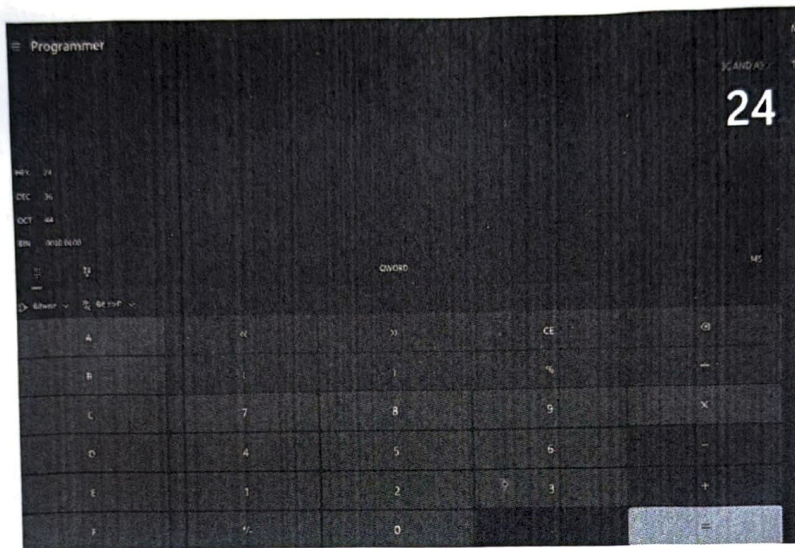
Objective

To perform AND, OR, XOR, and NOT operations on hexadecimal values.

1 – AND Operation

Explanation

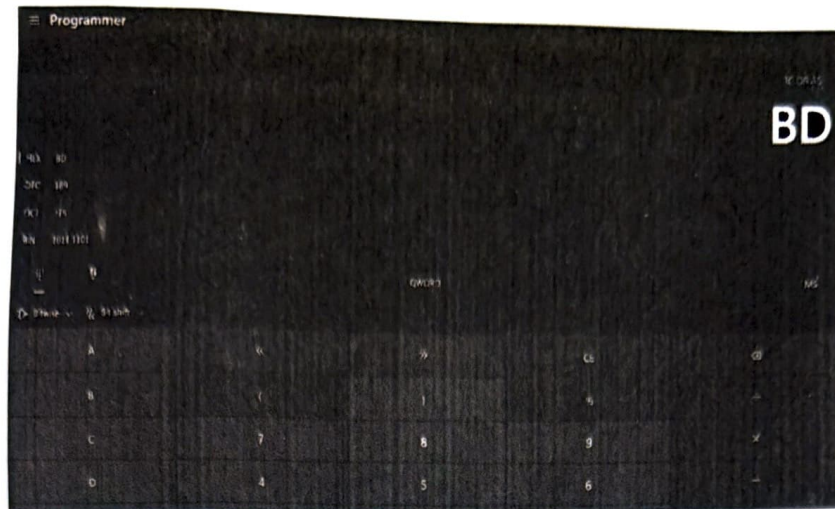
- AND gate gives **1 only when both input bits are 1**.
- Used for masking specific bits.



2 – OR Operation

Explanation

- OR gate gives **1 if at least one input bit is 1**.
- Used to set particular bits.



3 – XOR Operation

Explanation

- OR gate gives **1** if at least one input bit is 1.
- Used to set particular bits.



4 – NOT Operation

Explanation

- NOT gate inverts every bit.
- 0 becomes 1 and 1 becomes 0.

Q3. CPU, Memory and I/O Analysis

Task Manager

Type a name, publisher, or PID to search

Processes		32%	67%	2%	0%
Name	Status	CPU	Memory	Disk	Network
Apps (5)					
> Google Chrome (18)		0.1%	956.4 MB	0.1 MB/s	0 Mbps
> Microsoft Edge (10)		0.3%	299.9 MB	0.1 MB/s	0.1 Mbps
> Microsoft Word		0.1%	83.2 MB	0 MB/s	0 Mbps
> Notepad.exe		0%	83.8 MB	0 MB/s	0 Mbps
> Task Manager		0.6%	90.7 MB	0 MB/s	0 Mbps
Background processes (119)					
> AISpeech Audio Effect Control...		0%	0.9 MB	0 MB/s	0 Mbps

Observation:

- Notepad uses minimal CPU and memory.
- Chrome consumes higher CPU and RAM due to web browsing.
- VLC uses CPU and memory for video playback.
- The CPU, RAM, storage, and I/O devices work together to execute multiple applications efficiently.